

A scalar-valued positive subcase of the Akemann–Shell–Weaver projection problem

Source. The source paper is C. A. Akemann, G. C. Shell, and N. Weaver, *Locally nonconical convexity*, arXiv:0005194. Near the end of Section 4 the authors ask:

Let Q be a closed, bounded, strictly convex subset of a Hilbert space X
and let $T : X \rightarrow \mathbb{R}^n$ be a bounded linear map.
Is $T|_Q : Q \rightarrow T(Q)$ necessarily open?

The following proves the scalar case $n = 1$. The proof does not use closedness, strict convexity, or the Hilbert structure.

Theorem 1. *Let Q be a bounded convex subset of a normed linear space X , and let $T : X \rightarrow \mathbb{R}$ be a linear map. Then the restriction*

$$f = T|_Q : Q \longrightarrow T(Q)$$

is open, where Q and $T(Q)$ carry their relative norm topologies.

Proof. Fix $q \in Q$ and put $y = Tq$. Let U be a relative neighborhood of q in Q . It is enough to consider neighborhoods of the form

$$U_\varepsilon = Q \cap B(q, \varepsilon).$$

Since Q is bounded, choose $D > 0$ such that $\|x - q\| \leq D$ for every $x \in Q$. Fix $\alpha \in (0, 1]$ with $\alpha D < \varepsilon$.

For every $x \in Q$, convexity gives

$$q + \alpha(x - q) = (1 - \alpha)q + \alpha x \in Q$$

and the norm estimate gives this point in U_ε . Applying T therefore yields

$$(1 - \alpha)y + \alpha T(Q) \subseteq T(U_\varepsilon).$$

The set $T(Q)$ is a convex subset of $\mathbb{R}$, hence an interval possibly open, closed, half-open, degenerate, or unbounded only if Q is unbounded. Because Q is bounded, this interval is bounded. For every point y of an interval I , the homothetic interval $(1 - \alpha)y + \alpha I$ is a relative neighborhood of y in I : if y is an endpoint it contains a one-sided relative interval, if y is interior it contains a two-sided interval, and if $I = \{y\}$ this is trivial. Thus $T(U_\varepsilon)$ is a relative neighborhood of y in $T(Q)$. Since q and ε were arbitrary, f is open. $\square$

Corollary 1. *The Akemann–Shell–Weaver open problem has an affirmative answer when $n = 1$. In this scalar case the hypotheses that Q be closed, strictly convex, and Hilbert-space-valued are unnecessary; bounded convexity is sufficient.*

Remark 1 (Finite-dimensional reduction). *The same first paragraph proves a useful local criterion for arbitrary finite-dimensional targets. Let $C = T(Q)$, $y = Tq$, and $\varepsilon > 0$. If $D = \sup_{x \in Q} \|x - q\|$ and $\alpha D < \varepsilon$, then*

$$(1 - \alpha)y + \alpha C \subseteq T(Q \cap B(q, \varepsilon)).$$

Consequently $T|_Q$ is open at q whenever, for some such $\alpha > 0$, the homothetic copy $(1 - \alpha)y + \alpha C$ is a relative neighborhood of y in C . In particular the remaining obstruction for $n \geq 2$ can only occur at image points where $T(Q)$ has a genuinely non-conical local boundary, such as a cusp.

Bounded novelty check. The run indexes were searched for arXiv id 0005194, for the exact phrases “closed, bounded, strictly convex”, “ $T|_Q$ ”, and “cofinite-dimensional kernel”, and for title/author keywords from the source paper. No prior run packet or attempt was found for this target. A bounded local parsed-source search and exact-phrase web search found the source problem but no separate later explicit answer. This packet therefore records a partial positive subcase, not a claimed full answer to the $n \geq 2$ problem.